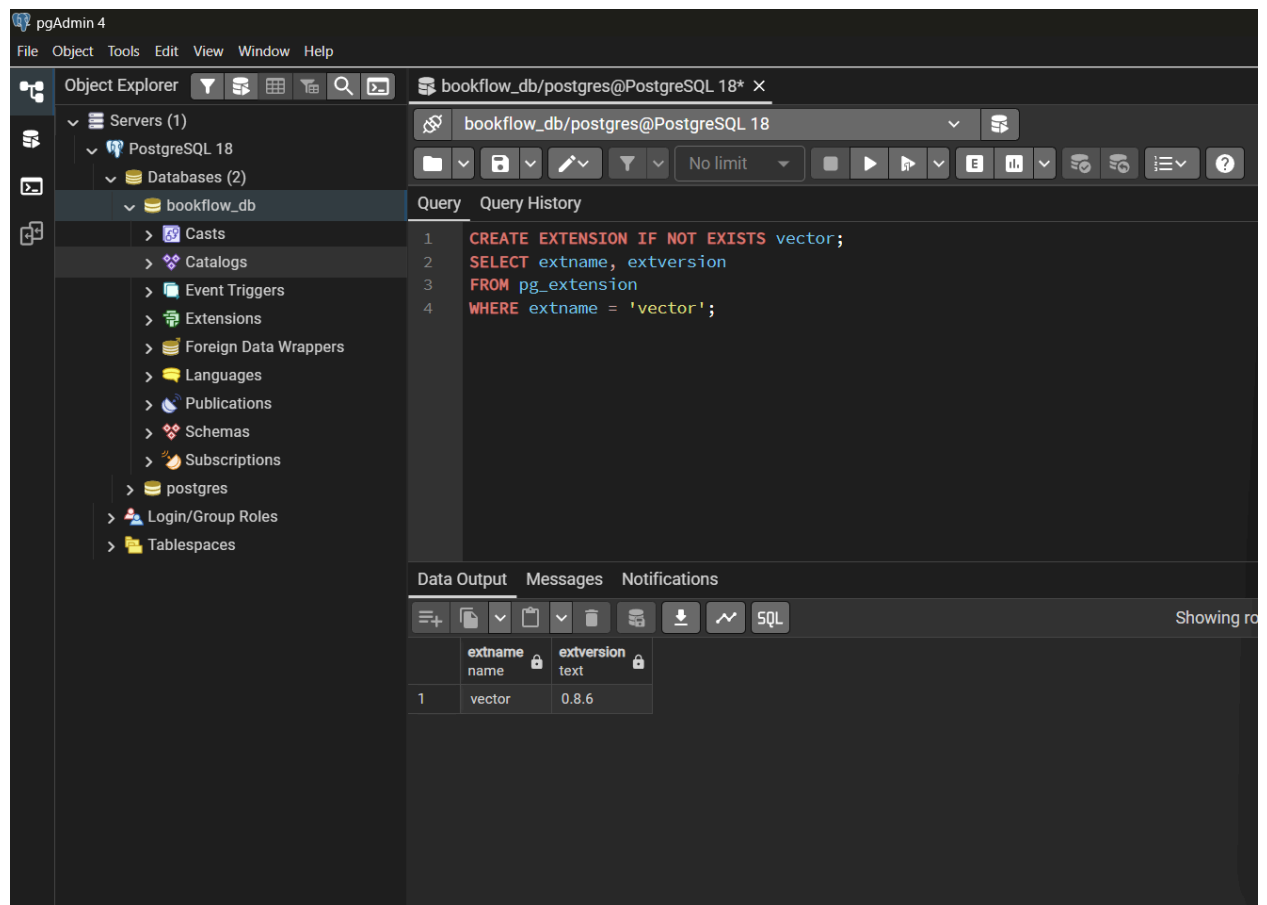
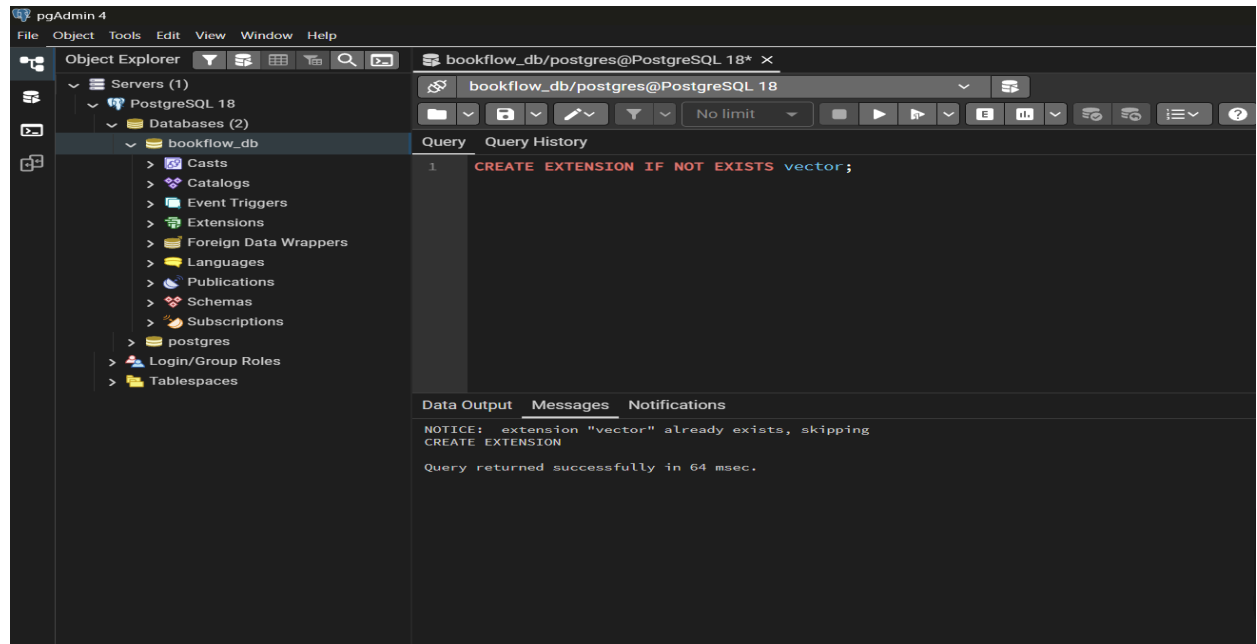
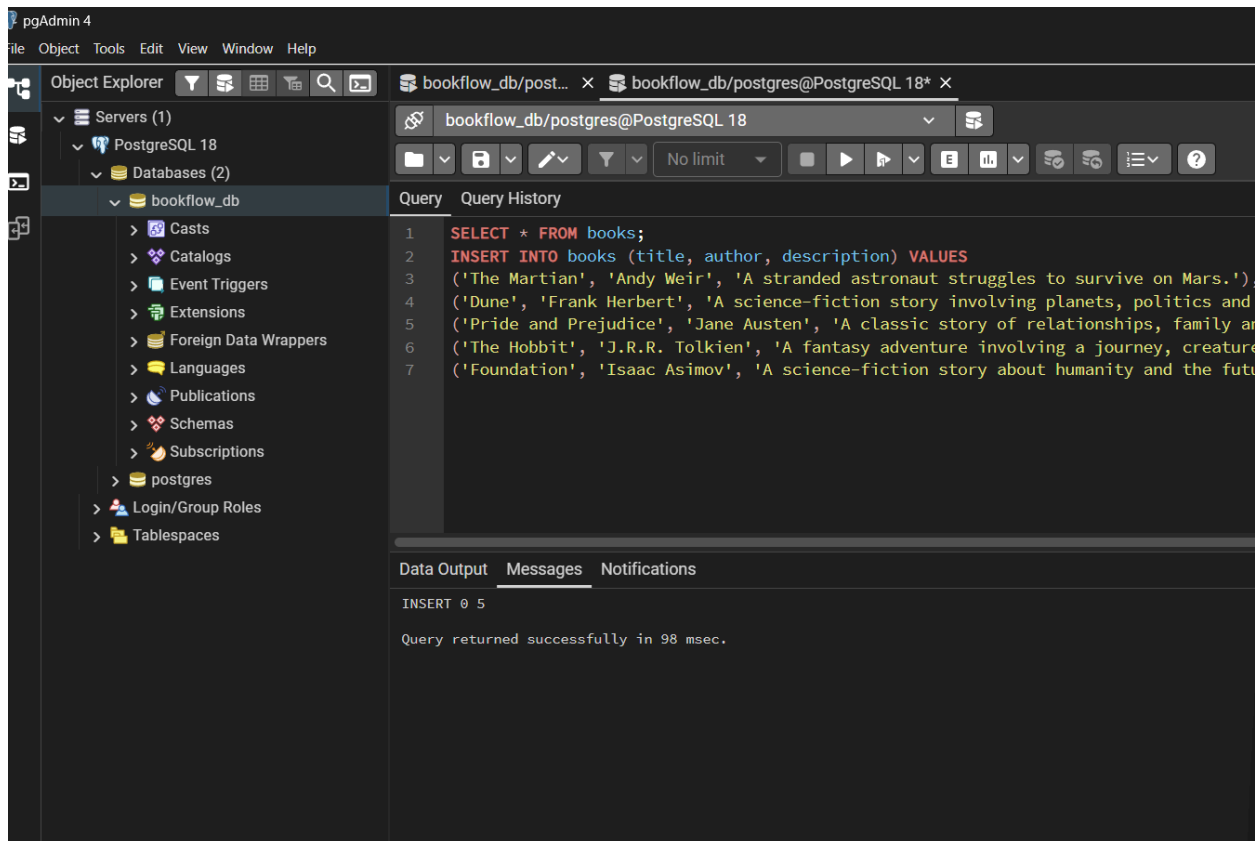
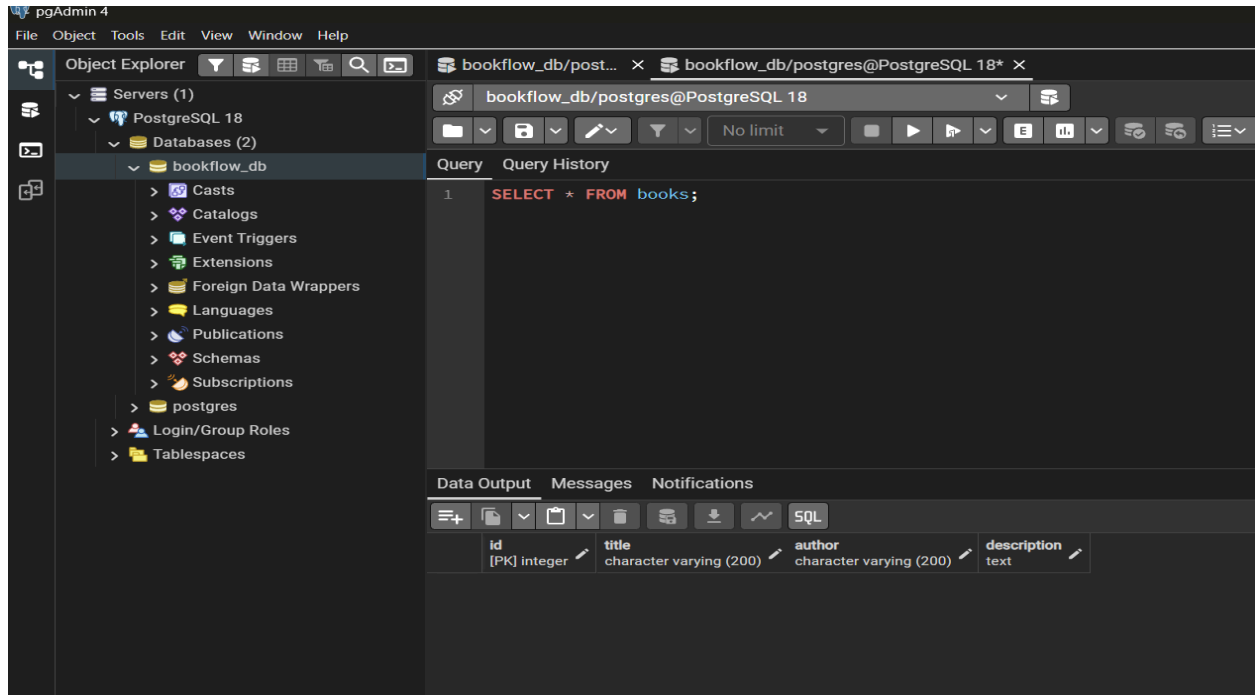






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bookflow_db/postgres@PostgreSQL 18

Query

```
1 SELECT * FROM books;
2 INSERT INTO books (title, author, description) VALUES
3 ('The Martian', 'Andy Weir', 'A stranded astronaut struggles to survive on Ma
4 ('Dune', 'Frank Herbert', 'A science-fiction story involving planets, politic
5 ('Pride and Prejudice', 'Jane Austen', 'A classic story of relationships, fam
6 ('The Hobbit', 'J.R.R. Tolkien', 'A fantasy adventure involving a journey, cr
7 ('Foundation', 'Isaac Asimov', 'A science-fiction story about humanity and th
8
9
10 SELECT column_name, data_type
11 FROM information_schema.columns
12 WHERE table_name = 'books';
```

Query History

Data Output Messages Notifications

	column_name	data_type
1	id	integer
2	title	character varying
3	author	character varying
4	description	text

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bookflow_db/postgres@PostgreSQL 18

Query

```
10 SELECT column_name, data_type
11 FROM information_schema.columns
12 WHERE table_name = 'books';
13 ALTER TABLE books ADD COLUMN embedding VECTOR(3);
14 UPDATE books
15 SET embedding = CASE title
16 WHEN 'The Martian' THEN '[0.10, 0.90, 0.20]':vector
17 WHEN 'Dune' THEN '[0.20, 0.85, 0.25]':vector
18 WHEN 'Pride and Prejudice' THEN '[0.80, 0.10, 0.70]':vector
19 WHEN 'The Hobbit' THEN '[0.30, 0.20, 0.90]':vector
20 WHEN 'Foundation' THEN '[0.15, 0.80, 0.30]':vector
21 END;
22 SELECT id, title, embedding
23 FROM books;
```

Query History

Data Output Messages Notifications

	id	title	embedding
1	1	The Martian	[0.1,0.9,0.2]
2	2	Dune	[0.2,0.85,0.2...
3	3	Pride and Prejudice	[0.8,0.1,0.7]
4	4	The Hobbit	[0.3,0.2,0.9]
5	5	Foundation	[0.15,0.8,0.3]
6	6	The Martian	[0.1,0.9,0.2]
7	7	Dune	[0.2,0.85,0.2...
8	8	Pride and Prejudice	[0.8,0.1,0.7]
9	9	The Hobbit	[0.3,0.2,0.9]
10	10	Foundation	[0.15,0.8,0.3]

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bookflow_db/postgres@PostgreSQL 18*

bookflow_db/postgres@PostgreSQL 18

Query Query History

```
END;  
SELECT id, title, embedding  
FROM books;  
  
SELECT  
    id,  
    title,  
    embedding <-> '[0.10, 0.85, 0.20]':vector AS l2_distance  
FROM books  
WHERE embedding IS NOT NULL  
ORDER BY embedding <-> '[0.10, 0.85, 0.20]':vector  
LIMIT 3;
```

Data Output Messages Notifications

	id [PK] integer	title character varying (200)	l2_distance double precision
1	1	The Martian	0.049999952875055514
2	6	The Martian	0.049999952875055514
3	2	Dune	0.11180339970798972

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bookflow_db/postgres@PostgreSQL 18

Query Query History

```
FROM books  
WHERE embedding IS NOT NULL  
ORDER BY embedding <-> '[0.10, 0.85, 0.20]':vector  
LIMIT 3;  
  
UPDATE books  
SET embedding = CASE title  
    WHEN 'The Martian' THEN '[0.10, 0.90, 0.20]':vector  
    WHEN 'Dune' THEN '[0.20, 0.85, 0.25]':vector  
    WHEN 'Pride and Prejudice' THEN '[0.80, 0.10, 0.70]':vector  
    WHEN 'The Hobbit' THEN '[0.30, 0.20, 0.90]':vector  
    WHEN 'Foundation' THEN '[0.15, 0.80, 0.30]':vector  
END;
```

Data Output Messages Notifications

UPDATE 10

Query returned successfully in 116 msec.

Object Explorer

Servers (1)

- PostgreSQL 18
 - Databases (2)
 - bookflow_db

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Query

```
31 LIMIT 3;  
32  
33 UPDATE books  
34 SET embedding = CASE title  
35 WHEN 'The Martian' THEN '[0.10, 0.90, 0.20]':vector  
36 WHEN 'Dune' THEN '[0.20, 0.85, 0.25]':vector  
37 WHEN 'Pride and Prejudice' THEN '[0.80, 0.10, 0.70]':vector  
38 WHEN 'The Hobbit' THEN '[0.30, 0.20, 0.90]':vector  
39 WHEN 'Foundation' THEN '[0.15, 0.80, 0.30]':vector  
40 END;  
41  
42 SELECT id, title, embedding FROM books;
```

Data Output

	id [PK] integer	title character varying (200)	embedding vector
1	1	The Martian	[0.1,0.9,0.2]
2	2	Dune	[0.2,0.85,0.2...
3	3	Pride and Prejudice	[0.8,0.1,0.7]
4	4	The Hobbit	[0.3,0.2,0.9]
5	5	Foundation	[0.15,0.8,0.3]
6	6	The Martian	[0.1,0.9,0.2]
7	7	Dune	[0.2,0.85,0.2...
8	8	Pride and Prejudice	[0.8,0.1,0.7]
9	9	The Hobbit	[0.3,0.2,0.9]
10	10	Foundation	[0.15,0.8,0.3]

Object Explorer

Servers (1)

- PostgreSQL 18
 - Databases (2)
 - bookflow_db

Schema Diff Workspace

bookflow_db/postgres@PostgreSQL 18*

Query

```
40 END;  
41  
42 SELECT id, title, embedding FROM books;  
43  
44 SELECT  
45 id,  
46 title,  
47 embedding <-> '[0.10, 0.85, 0.20]':vector AS l2_distance  
48 FROM books  
49 WHERE embedding IS NOT NULL  
50 ORDER BY embedding <-> '[0.10, 0.85, 0.20]':vector  
51 LIMIT 3;
```

Data Output

	id [PK] integer	title character varying (200)	l2_distance double precision
1	1	The Martian	0.049999952875055514
2	6	The Martian	0.049999952875055514
3	2	Dune	0.11180339970798972

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Query

```
49 WHERE embedding IS NOT NULL
50 ORDER BY embedding <=> '[0.10, 0.85, 0.20]':vector
51 LIMIT 3;
52
53 SELECT
54     id,
55     title,
56     embedding <=> '[0.10, 0.85, 0.20]':vector AS cosine_distance
57 FROM books
58 WHERE embedding IS NOT NULL
59 ORDER BY embedding <=> '[0.10, 0.85, 0.20]':vector
60 LIMIT 3;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	id [PK] integer	title character varying (200)	cosine_distance double precision
1	1	The Martian	9.408262185794936e-05
2	6	The Martian	9.408262185794936e-05
3	2	Dune	0.0072885302161147525

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Query

```
59 ORDER BY embedding <=> '[0.10, 0.85, 0.20]':vector
60 LIMIT 3;
61
62 SELECT
63     id,
64     title,
65     1 - (embedding <=> '[0.10, 0.85, 0.20]':vector) AS cosine_similarity
66 FROM books
67 WHERE embedding IS NOT NULL
68 ORDER BY embedding <=> '[0.10, 0.85, 0.20]':vector
69 LIMIT 3;
70
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	id [PK] integer	title character varying (200)	cosine_similarity double precision
1	1	The Martian	0.999905917378142
2	6	The Martian	0.999905917378142
3	2	Dune	0.9927114697838852

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```
1 openapi: 3.0.3
2 info:
3   title: BookFlow Search API
4   version: 1.0.0
5   description: API contract for semantic book search
6
7 paths:
8   /v1/books/search:
9     get:
10       summary: Search books semantically
11       operationId: searchBooks
12       parameters:
13         - name: q
14           in: query
15           required: true
16           description: Natural-language search query
17           schema:
18             type: string
19             minLength: 1
20         - name: limit
21           in: query
22           required: false
23           description: Maximum number of results
24           schema:
25             type: integer
26             minimum: 1
27             maximum: 20
28             default: 3
29       responses:
30         '200':
31           description: Search results
32           content:
```

API contract for semantic book search

default

GET /v1/books/search Search books semantically

Try it out

Parameters

Name	Description
q required	Natural-language search query
string (query)	q
limit	Maximum number of results
integer (query)	Default value : 3
	3

Responses

Curl

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```
1 openapi: 3.0.3
2 info:
3   title: BookFlow Search API
4   version: 1.0.0
5   description: API contract for semantic book search
6
7 paths:
8   /v1/books/search:
9     get:
10       summary: Search books semantically
11       operationId: searchBooks
12       parameters:
13         - name: q
14           in: query
15           required: true
16           description: Natural-language search query
17           schema:
18             type: string
19             minLength: 1
20         - name: limit
21           in: query
22           required: false
23           description: Maximum number of results
24           schema:
25             type: integer
26             minimum: 1
27             maximum: 20
28             default: 3
29       responses:
30         '200':
31           description: Search results
32           content:
```

Responses

Code	Description	Links
200	Search results	No links
Media type: application/json		
Controls Accept header		
Example Value Schema		
<pre>{ "query": "string", "results": [{ "id": 0, "title": "string", "score": 0 }] }</pre>		
400	Bad Request	No links
Media type: application/json		
Example Value Schema		
<pre>{ "error": { "code": "string", "message": "string" } }</pre>		